THE STALK OF A PROPER PUSHFORWARD OF A SHEAF OF SETS IS THE SECTIONS OVER THE GEOMETRIC FIBRE

THE STACKS PROJECT AUTHORS

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Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the computation of the stalk through the strict henselisation and the application of the h^0 comparison over a henselian local base.

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Allowed standard prerequisites (editorial).

- Stalks of pushforwards as sections over the strict henselisation. Same chapter, the theorem describing the stalk of a pushforward as sections over the base change along the strict henselisation. Supplies the identification of the stalk (f_*F)_s with sections over X x_S Spec(O^sh_{S,s}).
- The h^0 comparison over a strictly henselian local base. Same chapter, the lemma identifying sections over a proper scheme over a henselian local ring with sections over the closed fibre. Supplies the identification of those sections with Gamma(X_s, F_s).

Source reference. Original reference 0A3S: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18215–18215. Prerequisite pointer invoked at proof line(s) 18278; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03Q9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8015–8015. Prerequisite pointer invoked at proof line(s) 18267; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to S$ be a proper morphism of schemes. Let $\overline{s} \to S$ be a geometric point. For any sheaf $\mathcal{F}$ on $X_{étale}$ the canonical map*

$$(f_*\mathcal{F})_{\overline{s}} \longrightarrow \Gamma(X_{\overline{s}}, \mathcal{F}_{\overline{s}})$$

is bijective.

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Proof. By Theorem 03Q9 (for sheaves of sets) we have

$$(f_*\mathcal{F})_{\overline{s}} = \Gamma(X \times_S \mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}}), p^{-1}_{small}\mathcal{F})$$

where $p : X \times_S \mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}}) \to X$ is the projection. Since the residue field of the strictly henselian local ring $\mathcal{O}^{sh}_{S,\overline{s}}$ is $\kappa(s)^{sep}$ we conclude from the discussion above the lemma and Lemma 0A3S. $\square$